

PX4 SITL + Gazebo Classic + ROS 2 + RViz2

Integration Report

1. Objective

The objective of this task was to design and simulate a realistic UAV environment using **PX4 SITL with Gazebo Classic**, customize a PX4-compatible drone by adding a **payload box and a LiDAR sensor**, and visualize the LiDAR data in **ROS 2 using RViz2**. The task involved integrating multiple robotics tools into a single working simulation pipeline.

2. Software and Tools Used

- PX4 Autopilot (SITL)
- Gazebo Classic
- ROS 2 Humble
- RViz2
- Ubuntu Linux

All components were executed locally on the same machine and communicated through standard PX4–Gazebo and Gazebo–ROS interfaces.

3. Custom Gazebo World Creation

3.1 Purpose

To simulate a realistic environment containing obstacles that can be detected by the drone's LiDAR sensor.

3.2 Implementation

- A new Gazebo Classic world file was created.
- Multiple 3D models were added, including:
 - Buildings (cuboids)
 - Trees (cylinders and spheres)
 - Ground plane
- Objects were carefully:
 - Scaled to realistic proportions
 - Positioned at varying distances and orientations
 - Set as static models to avoid unintended motion

3.3 Result

A structured environment with surrounding obstacles that produces meaningful LiDAR returns during simulation.

4. Drone Model Customization

4.1 Base Model

The **Iris quadrotor**, which is PX4-compatible, was used as the base UAV model.

4.2 Payload Box Integration

Purpose

To simulate a physical payload attached beneath the drone.

Method

- The drone's SDF file was modified.
 - A new link representing a box was added:
 - Positioned below the drone's base link
 - Visual color set to blue for clarity
 - Physical properties were defined:
 - Mass
 - Inertia matrix
 - The payload was rigidly attached using a fixed joint.
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4.3 LiDAR Sensor Integration

Purpose

To enable environment sensing and ROS 2 visualization.

Method

- A LiDAR sensor was added as a separate link mounted on top of the drone.
- The sensor was defined using a ray-based LiDAR model.
- Key parameters configured:
 - Horizontal field of view (~360°)
 - Range limits
 - Angular resolution
 - Update rate
- A Gazebo ROS LiDAR plugin was used to publish data as `sensor_msgs/LaserScan`.

- The sensor frame was named `lidar_link`.
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5. PX4 SITL Integration

Purpose

To allow PX4 to control the drone while Gazebo handles physics and sensor simulation.

Process

- PX4 SITL was launched with the custom Iris model and custom Gazebo world.
- PX4 successfully connected to Gazebo and initialized:
 - Sensors
 - Parameters
 - Autostart configuration

Note

Interaction with the PX4 shell (`pxh>`) is **not required** for ROS 2 or RViz2 visualization. PX4 only needs to be running and connected to Gazebo.

6. ROS 2 Integration

Purpose

To access LiDAR data outside Gazebo and enable visualization.

Steps

1. ROS 2 Humble environment was sourced.
2. Active topics were listed to verify communication.
3. LiDAR topic `/lidar/scan` was identified.
4. Data validity was confirmed using a single message echo.

Observation

- Messages of type `sensor_msgs/LaserScan` were published correctly.
 - Range values changed dynamically based on nearby obstacles.
 - The `frame_id` was correctly set to `lidar_link`.
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7. RViz2 Visualization

Purpose

To visually confirm LiDAR data and validate frame alignment.

Steps

- RViz2 was launched in a new terminal.
- Fixed Frame was set to `lidar_link`.
- Displays added:
 - Grid
 - LaserScan
- LaserScan display was configured with:
 - Topic: `/lidar/scan`
 - Appropriate point size and style

Result

Live LiDAR points were visible in RViz2 and matched the obstacle layout seen in Gazebo.

8. Validation and Debugging

Verification Methods

- Compared Gazebo LiDAR rays with RViz2 point visualization.
- Confirmed ROS 2 topic output matched visual data.
- Ensured no blocking TF errors affected visualization.

Confirmed Data Flow

PX4 → Gazebo → LiDAR Plugin → ROS 2 → RViz2

9. Final Outcome

- A custom Gazebo world with obstacles was successfully created.
- The Iris drone was modified to include a payload and LiDAR sensor.
- PX4 SITL controlled the drone within the simulated environment.
- LiDAR data was published to ROS 2 and visualized in RViz2.

All task deliverables were completed and validated using both terminal output and graphical visualization.

10. Summary Explanation (For Viva or Review)

The task was completed by customizing the Iris UAV model to include a payload and a LiDAR sensor, creating a realistic Gazebo Classic world, running the simulation using PX4 SITL, bridging the LiDAR data into ROS 2, and finally visualizing the LaserScan output in RViz2 using proper frame and topic configuration.